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Experiment - 1

Study of various Network commands used in Linux and Windows

Aim:
To study of various network commands used in Linux and windows

Windows commands

* arp -a
ARP is short form of address resolution protocol, it will show IP address of your computer along with IP address and MAC address of your router.

* hostname
This is the simplest of all TCP/IP commands. It displays name of your computer

* ipconfig /all:
This command displays detailed config information about your TCP/IP connection

* nbtstat -a:
This command helps solve problems with NetBios name resolution

Netstat:

netstat displays a variety of statistics about a computers active TCP/IP connections

· nslookup:

It is a tool used to perform DNS lookups in Linux. It is used to display DNS details. such as IP address of a particular computer.

pathping:

Pathping is unique to windows and is basically a combination of ping and Traceroute commands.

ping:

It is the best way to test connectivity between two nodes.

1. #ping hostname (ping localhost)
2. #ping ip address (ping 4.22.2)
3. # ping (ping www.facebook.com)

Route:

Route command is used to show/ manipulate the IP routing table.

Linux commands.

* ip

The ip command can show address information, manipulate routing, plus display network various devices, interfaces and tunnels

* [root@server~]# ip address show
  - to show the IP address assigned to an interface
* [root@server~]# ip address add 192.168.1.254/24 dev enp503
  - to assign a IP to an interface
* [root@server~] # ip address del 192.168.1.254/24 dev enp503

* ifconfig

The ifconfig command is a staple in many sysadmin's tool belt for configuring and troubleshooting networks.

* tntr

This command combines the functionality of the ping and traceroute commands.

* [root@server ~]# mtr google.com
  - the basic mtr command shows you the statistics

* [root@server ~]# mtr -g google.com
  shows numeric IP address

* [root@server ~]# mtr -b google.com
  shows numeric IP address and hostnames

* tcpdump
  It is designed for capturing and displaying packets

* [root@server]# dnf install -y tcpdump
* [root@server ~]# tcpdump -p

* for traffic coming from 8.8.8.8 use
  [root@server ~]# tcpdump -i eth0 -c10 host 8.8.8.8

* For outbound traffic going to 8.8.8.8 use
  [root@server ~]# tcpdump -i eth0 src host 8.8.8.8

* For specific host,

[root@server ~]# tcpdump -p eth0 host 8.8.8.8 and port 53

* To capture only HTTPS traffic,

[root@server ~]# tcpdump -p eth0 -c 10 host www.google.com and port 443.

* To capture only all port except port 80 and 25.

[root@server ~]# tcpdump -p eth0 port not 53 and not 25.

* pping.

pping is a tools that verifies IP-level connectivity to another TCP/IP computer by sending Internet control Messaging Protocal (ICMP) Echo Request messages

* [root@server ~]# pping gode.com

Pinging google.com (216.58.206.174) 56(84) bytes of data.

64 bytes from sof02s27-in-f14.1e100.net (216.58.206.14): Picmp-seq=1 ttl=56 time=10.4 ms

^c - to stop pping command (ctrl+c).

* To pping a host ten times, use the following command.

- [root@server ~]# pping -c 10 google.com.

Result:

The above commands are executed successfully.

02/08/26

Experiment-2

# Study of different types of Network cables

Aim: To study of different types of Network cables.

a) Understand different type of network cable

Different types of cables of network cables

1. Unshielded Twisted Pair (UTP) Cable
2. Shielded Twisted Pair (STP) Cable.
3. Coaxial cable.
4. Fibre Optic Cable.

| Cable Type | Category | Maximum Data Transmission | Advantages / Disadvantages | Application / Use. |
|---|---|---|---|---|
| UTP | Category 3 | 10 bps | Advantage<br>* cheaper in cost<br>+ Easy to install as they have smaller overall diameter<br>Disadvantage<br>+ More prone to (EMI) Electromagnetic Interference and noise | 10 Base-T Ethernet<br>Fast Ethernet<br>Gigabit Ethernet |
| | Category 5 | Up to 100 MBps | | |
| | Category 5e | 1Gbps | | Fast Ethernet)<br>Gigabit Ethernet |

| | | | | |
|---|---|---|---|---|
| STP | category 6; 6a | 10Gbps | Advantage<br>* shielded<br>+ Faster than UTP<br>+ Less susceptible to noise and interference | 10Gigabit Ethernet<br>10 or Ethernet (55m)<br>widely used in data centres |
| SSTP | Category 7 | 10Gbps | Disadvantage<br>* Expensive<br>* Greater installation effort | 10Gigabit Ethernet, 10 or Ethernet (100 m) |
| coaxial cable | RG-6<br>RG-59<br>RG-11 | 10 - 100 Mbps | Advantage<br>* High bandwidth<br>* Immune to interference<br>* Low loss bandwidth<br>* Versatile<br><br>Disadvantage<br>* Limited distance<br>* Cost<br>* Size & bulky | Speed of signals<br>500m<br>Television network<br>High speed internet connections |

| | | | | |
|---|---|---|---|---|
| fibre optics cable | single mode multi mode | 100Gbps | Advantage<br>* High speed<br>* High bandwidth<br>* High security<br>* Long distance<br>Disadvantage<br>* Expensive<br>* Requires skilled installers | Maximum distance of fibre optics cable is around 100 meters |

Result:

The above different types of cables are explained successfully.

4/8/24

Experiment-3

To study the Packet tracer tool Installation and User Interface

Aim:
To understand environment of CISCO PACKET TRACER To design simple network

Introduction:

As simulator, as the name suggests, simulates network devices and its environment

1. It allows you to model complex systems without the need for dedicated equipment

2. It helps you to practice your network configuration

3. It is available for the Linux and windows desktop environment

Installing Packet tracer

*CISCO Packet Tracer can be downloaded from the cisco networking Academy website.

*After downloading the appropriate version for OS, the Installation can be completed by following setup instructions.

User Interface overview

1. Menu Bar : Provides options.

2. Main Toolbar - Contains commonly used tools.

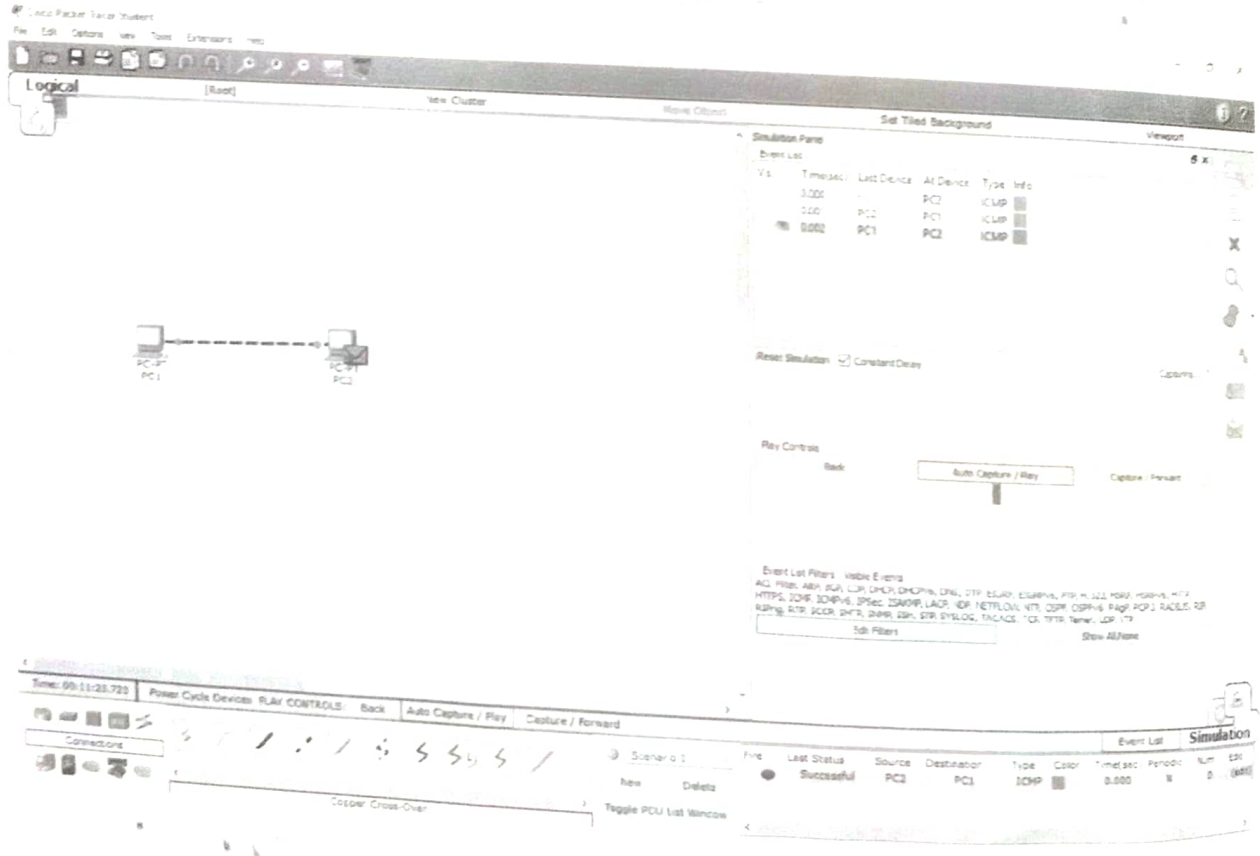
3. logical/physical workspace Tabs - used to switch logical and physical view

4. workplace - Area where network topologies are created.

5. common Tool Bars - Provides hook for

6. User-created Product Box - used create customized products

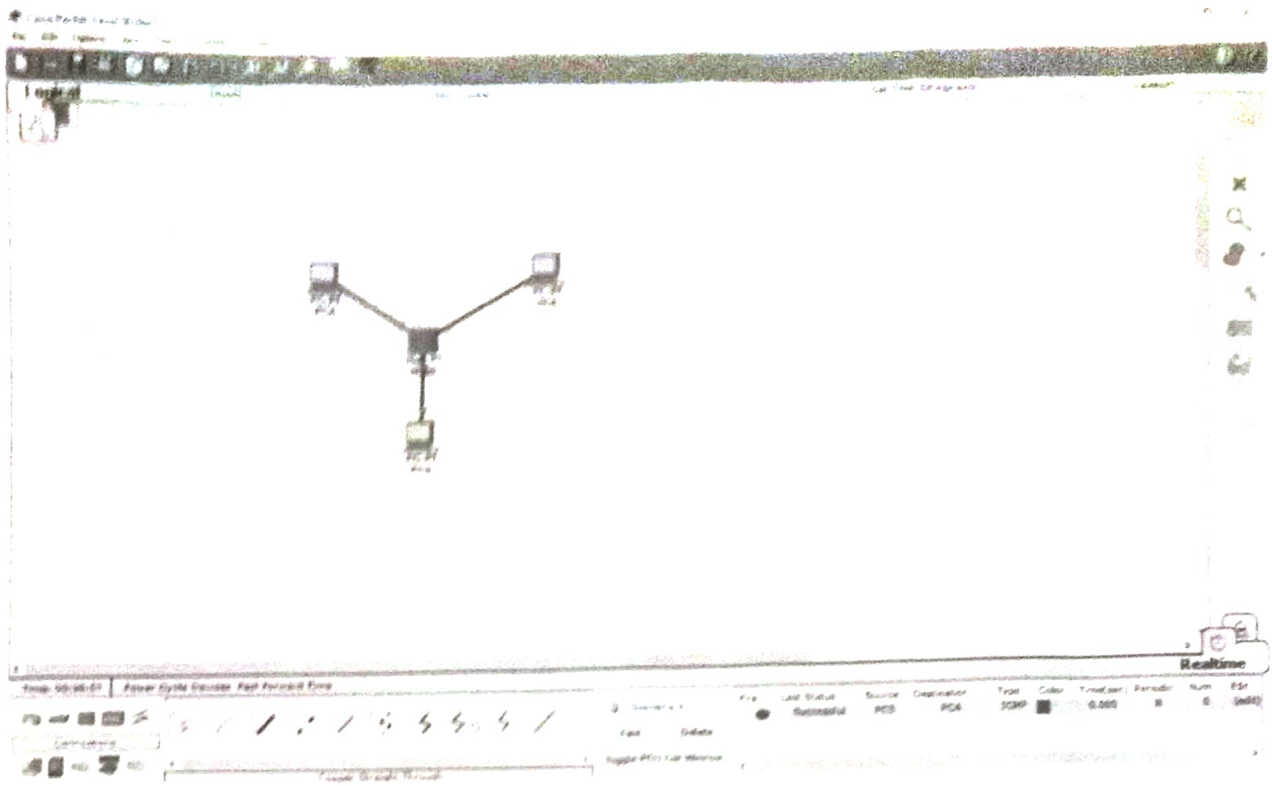
Peer to Peer




* Place two PC and connect them using cross over cable.

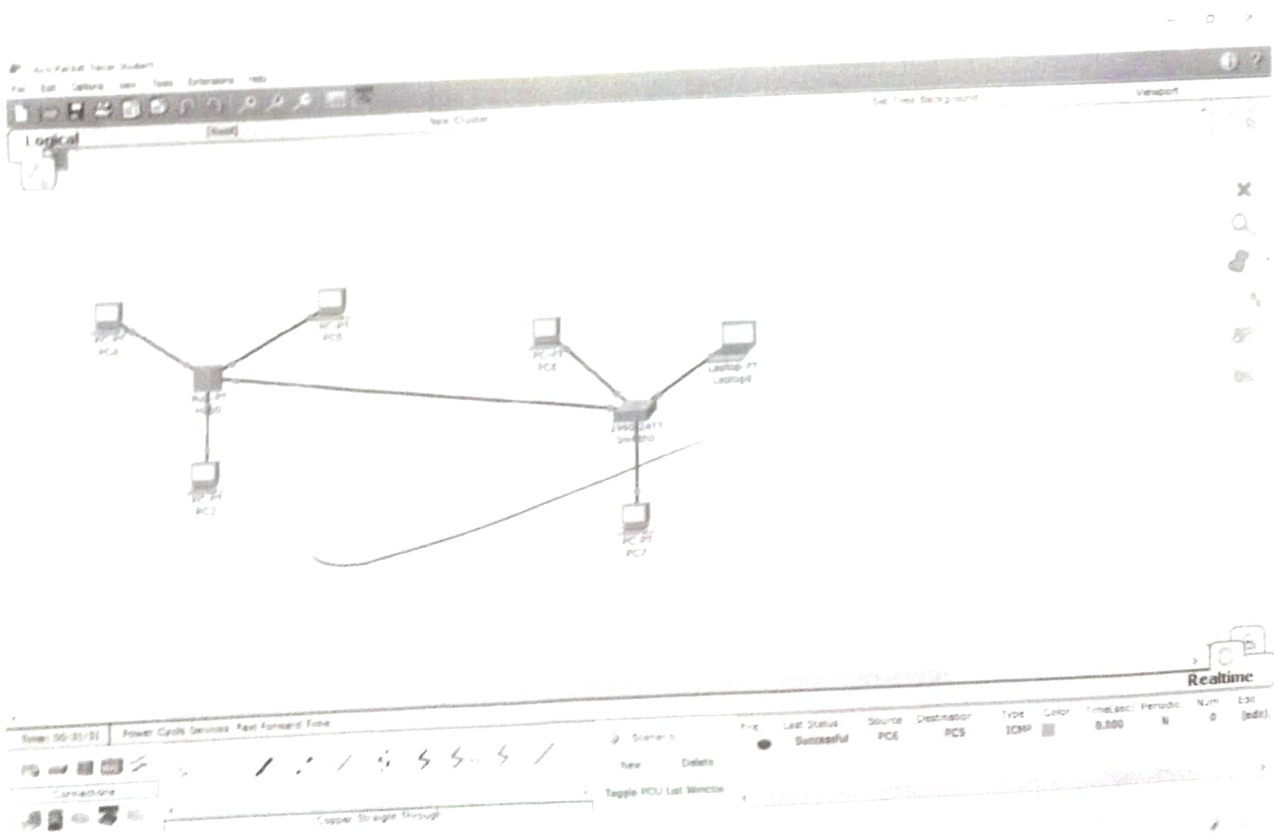
* configure IP address for both PC's in same network.

Hub Network



* Place the PCs and a Hub, then connect them using copper straight through cables.
* call. Ping to verify communication the PCC's

Switch Network



* Place a switch and the connect PT to the hub and other PC's using copper straight - Through cables.

* Configure IP address for the connection devices

IP Configuration

| Device | IP address | Subnet mask |
|---|---|---|
| PC0 | 10.1.1.1 | 255.0.0.0 |
| PC1 | 10.1.1.2 | 255.0.0.0 |

Result: Cisco Packet Tracer was successfully verified and used to create a simple network.

# Experiment - 4.

Setup and configure a LAN using a switch and Ethernet cables in your lab.

Aim: To setup and configure a LAN using a switch and Ethernet cables in your lab.

LAN - A Local Area Network refers to a network that connects devices within a limited area, such as an office building, school, or home.

Set up a LAN

Step 1: Plan and design an appropriate network topology and location.

Step 2: You can take 4 computers, switch and 4 Ethernet cables.

Step 3: Connect your computer to network switch via Ethernet cable.

Step 4: Assign IP address to your PC's.

Steps: configure a network switch

1. Connect your computer to the switch
2. Log in to the web interface.
3. Configure basic settings
4. Assign IP address

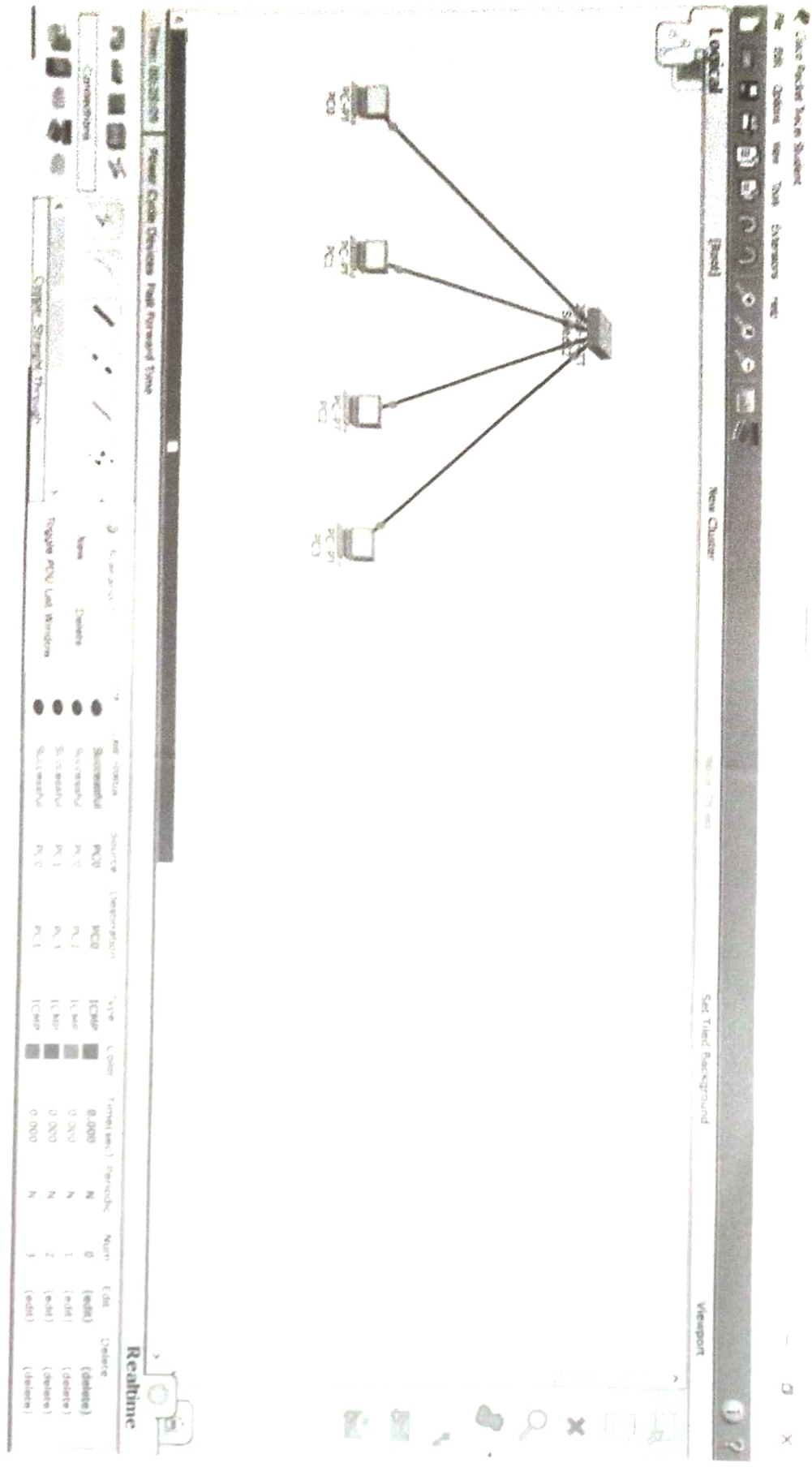
Step 6: check connectivity between switch and other machine by using ping command.

Step 7: Select a folder, -> go to properties -> click sharing tab -> share it with everyone on same LAN.

Step 8: Try to access the shared folder from other computers of the network

LAN connection

Diagram:



Result:

Thus the configuration of a LAN using a switch and Ethernet cables are sucessfully performed.

arp

| No. | Time | Source | Destination | Protocol | Length | Info |
|---|---|---|---|---|---|---|
| 6 | 0.255305 | FoxConn_c9:c5:40 | Broadcast | ARP | 60 | Who has 172.16.0.15? Tell 172.16.0.3 |
| 14 | 0.982936 | FoxConn_d8:4c:46 | Broadcast | ARP | 60 | Who has 172.16.0.3? Tell 172.16.0.8 |
| 19 | 1.418424 | FoxConn_c9:c5:91 | Broadcast | ARP | 60 | Who has 172.16.0.8? Tell 172.16.0.26 |
| 24 | 1.880729 | FoxConn_d8:4c:46 | Broadcast | ARP | 60 | Who has 172.16.0.1? Tell 172.16.0.8 |
| 27 | 2.029517 | 61:0a:92:d2:e4 | Broadcast | ARP | 60 | Who has 172.16.0.1? Tell 172.16.0.1 |
| 41 | 2.509905 | 61:0a:7c:c5:54 | Broadcast | ARP | 60 | Who has 172.16.0.1? Tell 172.16.0.131 |
| 44 | 2.602358 | FoxConn_c9:c8:24 | Broadcast | ARP | 60 | Who has 172.16.0.82? Tell 172.16.0.131 |
| 46 | 2.743021 | De:11:35:11:11:11 | Broadcast | ARP | 60 | Who has 172.16.0.179? Tell 172.16.0.22 |
| 56 | 3.201822 | 61:0a:92:d2:e4 | Broadcast | ARP | 60 | Who has 172.16.0.114? Tell 172.16.0.195 |
| 60 | 3.237061 | 61:0a:7c:c5:34 | Broadcast | ARP | 60 | Who has 172.16.0.94? Tell 172.16.0.1 |
| 71 | 3.670045 | De:11:35:11:11:11 | Broadcast | ARP | 60 | Who has 172.16.0.82? Tell 172.16.0.111 |

```
Frame 119: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface 0
Ethernet II, Src: IntelCor_13:ed:7c (00:27:0e:13:ed:7c), Dst: RealtekS_b2:60:90 (00:e0:4c:b2:60:90)
Address Resolution Protocol (reply)
```

```
0000  00 e0 4c b2 60 90 00 27  0e 13 ed 7c 08 06 00 01   ..L.`..'  ...|....
0010  08 00 06 04 00 02 00 27  0e 13 ed 7c [redacted]    .......'  ...|....
0020  00 e0 4c b2 60 90 0a 10  09 64                      ..L.`...  .d
```

# Experiment-5

## Experiment on Packet capture tool : Wireshark

Aim:
To experiments on Packet capture tool: Wireshark

Packet Sniffer.

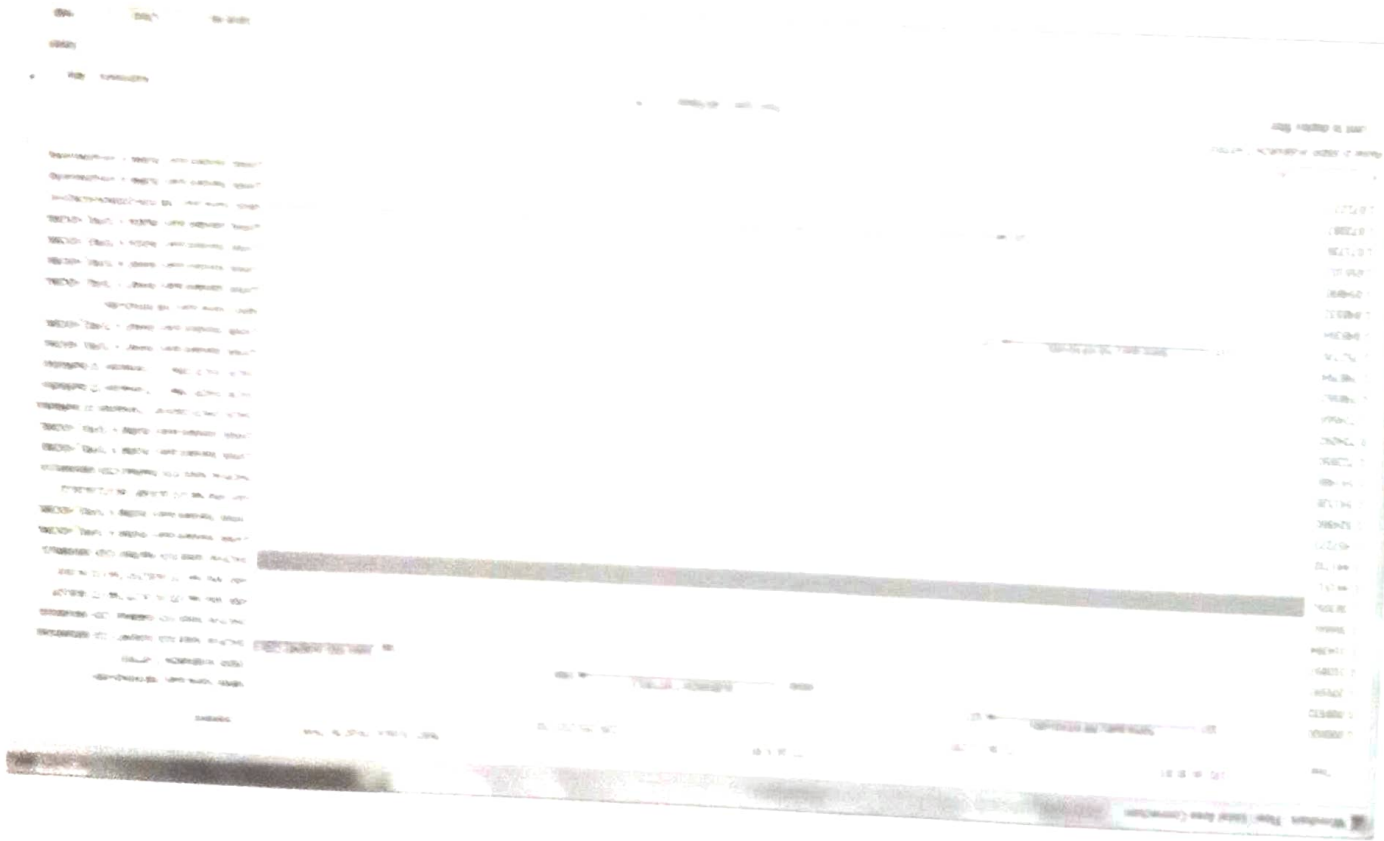
* Sniffs messages being sent/received from by your computer
* store and display the contents of various protocol fields in message
* passive program
  1) Never sends packet itself.
  2) No packets added to it
  3) Receives a copy of all packet (sent/receive)

Packet Sniffer structure; Diagnosis tcp tool:

* Top dump - Eg: sp dep one hat 10.12.90.42.w
* wireshark - wireshark -r exe.5 out

Procedure

* select the local Area network in wire shark



* Go to capture option
* Select capture and stop automatically after two packets
* Then click start capture
* Save packets

8) Create a filter to display only TCP / UDP packets, inspects the packets and provide flow graph

* Go to search bar at the top of the tool.
* Search TCP packets on search bar
* To see flow graph click statistics flow graph
* Save the packets

8a) Create a filter for ARP Packets

* Go to search option
* Search ARP packets on search bar
* Save the packets

iii) creating a filter to display DNS packets and provide the flow graph

iv) creating filter to display HTTP only packets and inspect the packet.

v) creating Filter for ICMP and DHCP packets and inspects and show the packets.

Result:

Wireshark helps to us in analyzing and understand our network traffic and packet flow throw given filters and flow graph

Exp-6

# PYTHON IMPLEMENTATION OF HAMMING CODE CONCEPT

Error correction of Data Link Layer :

Hamming code is a set of error-correction codes that can be used to detect and correct errors when data is transmitted from sender to receiver.

Aim:

To implement error detection using Hamming code Technique at Data Link Layer

SENDER Program

CODE:

```
def hamming_encode(data):
    encoded = ""
for p in range(0, len(data), 4):
    d = data[p:p+4]
    if len(d) < 4:
        d = d.ljust(4, '0')
    d1, d2, d3, d4 = map(int, d)
    p1 = d1 ^ d2 ^ d4
    p2 = d1 ^ d3 ^ d4
    p4 = d2 ^ d3 ^ d4
```

```
code = str(p1) + str(p2) + str(d1)
    + str(p4) + str(d2) + str(d3) +
    str(d4).

encode += code.
return encoded.
text = input("Enter text: ")
binary = ''.join(format(ord(i), '08b') for i in text)

print("Binary data", binary)
encoded_data = hamming_coded(binary)
with open("channel.txt", "w") as f:
    f.write(encoded_data)
print("Hamming encoded data:",
      encoded_data)
print("Data saved to channel.txt")
```

RECEIVER PROGRAM

CODE:

```
def hamming_decode(coded):
    corrected = ""
    error_found = False
```

```
for i in range(0, len(data), 7):
    block = list(map(int, data[i:i+7]))
    if len(block) < 7:
        continue
    p1, p2, d1, p4, d2, d3, d4 = block
    s1 = p1 ^ d1 ^ d2 ^ d4
    s2 = p2 ^ d1 ^ d3 ^ d4
    s4 = p4 ^ d2 ^ d3 ^ d4
    error_position = s1 + (s2 * 2) + (s4 * 4)
    if error_position != 0:
        error_found = True
        actual_position = i + error_position
        print("Error detected at bit position:", actual_position)
        block[error_position - 1] ^= 1
        print("Error corrected at bit position:",
              actual_position)
    corrected += str(block[2])
    corrected += str(block[4])
    corrected += str(block[5])
    corrected += str(block[6])
return corrected, error_found
with open("channel.txt", "r") as file:
```

```
channel_data = file.read(); strip()
print("Data received from channel:")
print(channel_data)
binary_data, error_found = hamming_decode(channel_data)
text = ""
for i in range(0, len(binary_data), 8):
    byte = binary_data[i:i+8]
    if len(byte) == 8:
        text += chr(int(byte, 2))
print("\n corrected binary data:")
print(binary_data)
if error_found:
    print("\n Error was detected and corrected.")
else:
    print("\n No error was detected")
print("Received text =", text)
```

INPUT:

Enter the text: HELLO

INPUT :

HELLO

OUTPUT :

Enter the text: HELLO

Error detected at position : 1

Error corrected.

original Message : HELLO

Result :

The above hamming code was executed successfully.

Exp: 7

# Flow control at Data Link Layer:

Aim

To write a program to implement flow control at data link layer using SLIDING WINDOW

Program should achieve atleast below given requirement you can make it a bidirectional program

Create a sender program with Flowing features:

1) Input window size from user
2) Input Text message from the user
3) Consider character per frame
4) Create a frame with following fields
5) Send the frames

[For the output on screen and save it in a file called sender Buffer]

6) Wait for the acknowledgment from the receiver
7) check ack. Reader a file called Receiver buffer
8) check Ack field the Acknowledgments number

9) If the Acknowledgement numbers is as expected send new set of frames accordingly.

Create a receiver file with following features:

1) Read a file called sender - Buffer
2) check frame no
3) If frame no are as expected write the appropriate Ack no in Receiver - Buffer file Else write NACK no in Receiver - Buffer file.

Sender code:

```
Import os
Import time
Base - DIR = os.path.dirname(os.
    path.abs path(__)
print("Sender Buffer", send.txt)
print("Receiver Buffer", Receiver.txt)
window - size = int(Input("Enter window
    size:"))
message = Input("Enter text message:")
```

```
if window_size <= 0:
        Print ("window size must be
            greater than zero")
        fake_systemsd;

Frames[]
    while start < total_frames:
        end = min(start + window_size,
                total_frames)
        Print ("\n = = = = = = = = ")
        print ("sending frames", start, "to", end)

elif ack_type == "NACK":
        print ("NACK received")
        print ("Retransmitting from frames:",
            ack_no).

else:
        print ("unknown error: Retrying current
                window: !!")

    Print ("Total Frames", total_frames).
```

Receiver Code:

```
import os
import time
expected_frame = 1
```

```
last_content = "4"
for line in lines:
try:
    parts = line.split(',')
except (IndexError, ValueError):
    break;
else:
    print("Frame error detected")
    print("Expected Frame:4",
          expected_frame)
    error_found = True
    print("waiting for next
          window...")
    time.sleep(0.5)
```

Sender code Input

window size : 3

Message : HELLO

Sender code Output

Frame 0 : H

Frame 1 : E

Frame 2 : E

Frame 3 : L

Frame 4 : O

Receiver code Input.

Sender . Buffer. txt .

0 H
1 E
2 L
3 L
4 O .

Receiver code Output

Received 0 : H

Received 1 : E

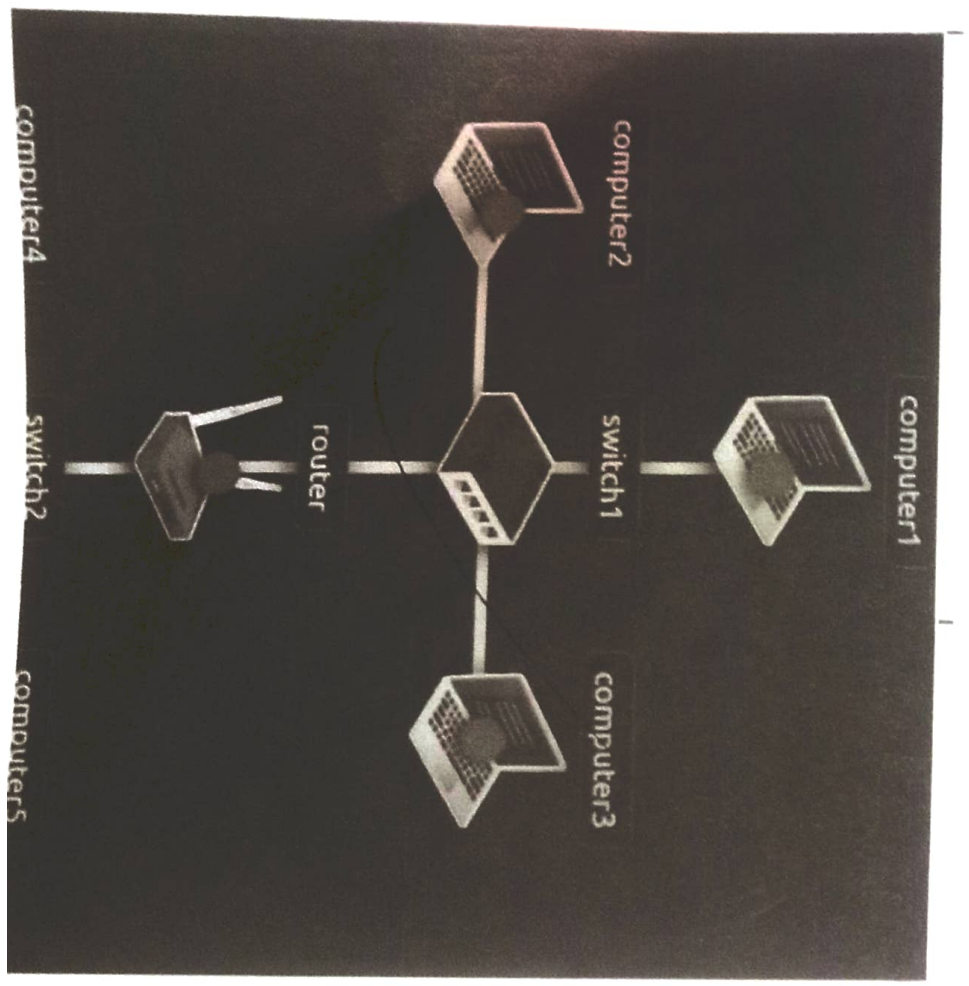
Received 2 : L

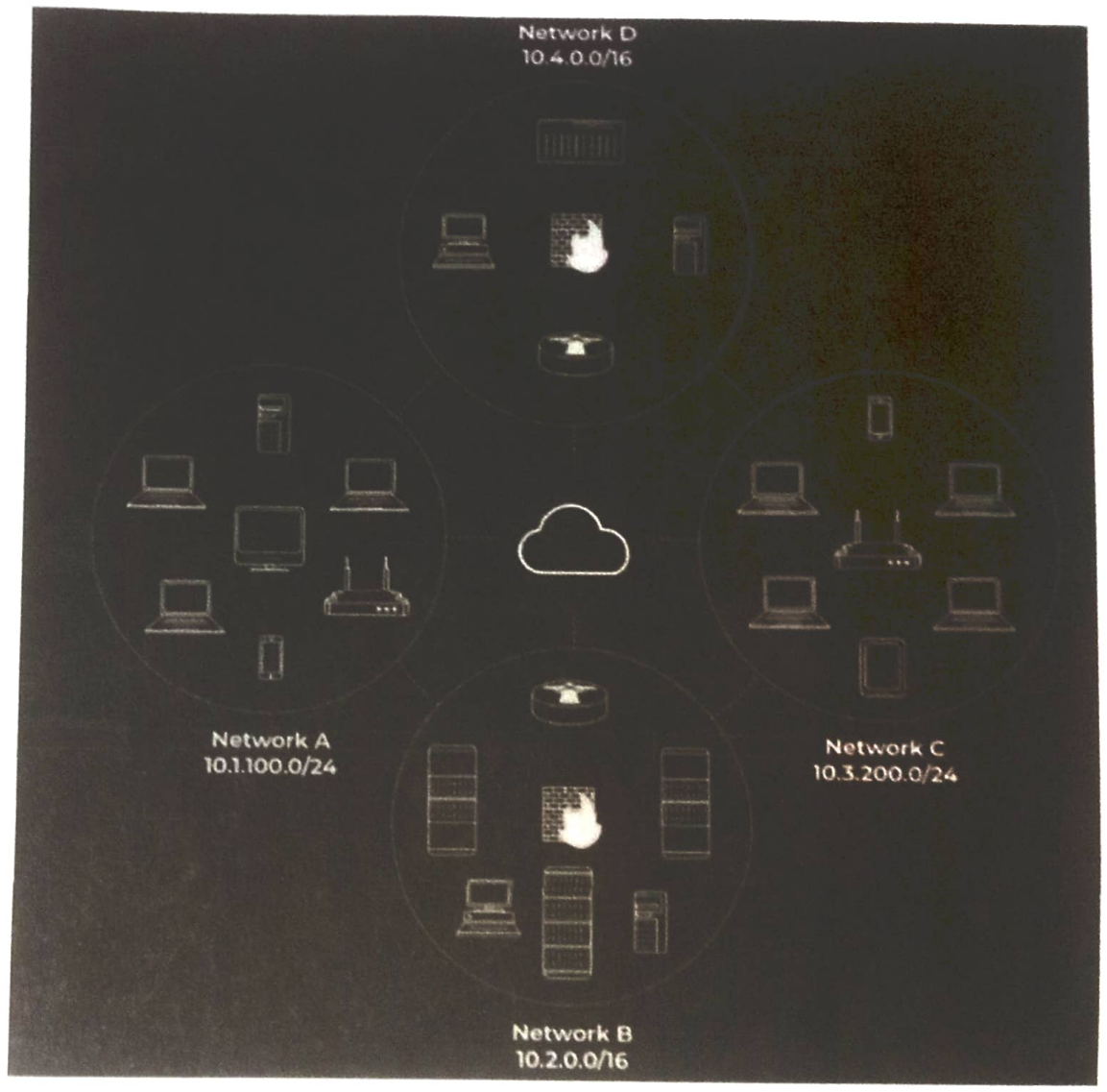
Received 3 : L

Received 4 ; O

Result:

Therefore, the Implement flow control of datalink layer using Slanding windows protocol executed successfully.


computer4
computer2
computer1
switch2
router
switch1
computer5
computer3


Network D
10.4.0.0/16
Network A
10.1.100.0/24
Network C
10.3.200.0/24
Network B
10.2.0.0/16

Exp: 8

NMAP to Discover Live Hosts Using Nmap Scans (ARP, ICMP, TCP/UDP) on TryHackMe Platform.

Aim:

To Discover Live Host using Nmap scan (ARP, ICMP, TCP/UDP) on the try hackme Platform Room Link:

https://tryhackme.com/room/nmap

Subnetworks:

A network segment is a group of computers linked through a shared medium like an Ethernet switch or coffee access point. In IP networks, a subnetwork typically consists as a logical connection.

In provided network diagram, there are four network segments or subnetwork, your systems would manually connected security policies

The figure displays two types of subnets

4 subnets. These have a subnet mask of 255.255.0.0 and can accomodate approximately

124 subnets These feature a subnets

key to run compiler

i) From computer 1

ii) To computer 1 (to include if its low)

iii) Packet Type = "ARP Request"

iv) Data computer

Discovering Live host.

TCP IP layers

i) ARP from Link layer

ii) ICMP From Network layer

iii) TCP From Transport Layer

iv) UDP From Transport Layer.

send a packet with following

i) From computer 1

ii) To computer 3

NMAP HOST Discovery using ARP.

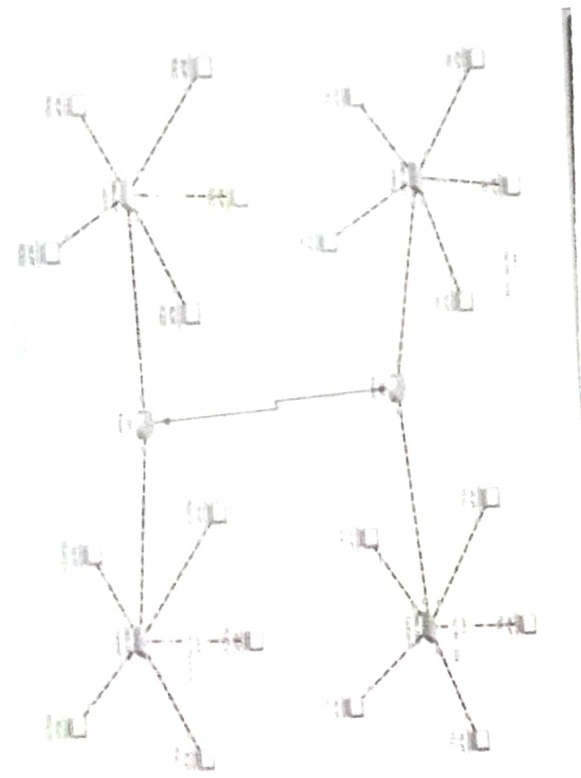
How can can determine which host are operational?

It is crucial to prevent unnecessary port scanning on host that are offline not in use.

This allows you to discover the system on same subnet as your target machine. Consider any port scans.

Result:

Therefore, the Discover Host using Nmap scan platform task was executed successfully.



Exp : 9.

Aim:

Implementation of Subnetting in cisco Packet Tracer Simulator.

Procedure

i) classless IP subnetting is a technique that allows for more efficient use of IP address by allowing for subnet masks networks.

ii) CREATING A NETWORK TOPOLOGY

The first step in implementing classless IP subnetting is to create a network topology in packet trace.

To create a network technique to add devices.

iii) Adding the Devices:

Once we have created our networking topology, we can add devices to it

iv) SUBNETTING:

To Subnet the network address of 192.168.1.0/24 to provide energy space for atleast 5.

The IP Addressing for the network shown on topology

i) Router R1

ii) GigabPt Element 0/0 : 192.168.1.1

iii) GigabPt Ethernet 0/1 : 192.168.2.1

iv) Switch S1

v) Fast Ethernet 0/1 : 192.168.1.0/27

vi) PC1 : 192.168.1.11

vii) PC2 : 192.168.1.12

viii) PC3 : 192.168.1.13

ix) PC4 : 192.168.1.14

x) PC5 : 192.168.1.15

xi) Fast Ethernet 0/2 : 192.168.2.0/27

xii) PC5 : 192.168.4.15

Configure the device

# enable

# configure terminal

# Interface Fast Ethernet 0/0

# Ip address {IP address} {subnet mask}

# no shutdown

# exit

Interface Fast Ethernet 0/1

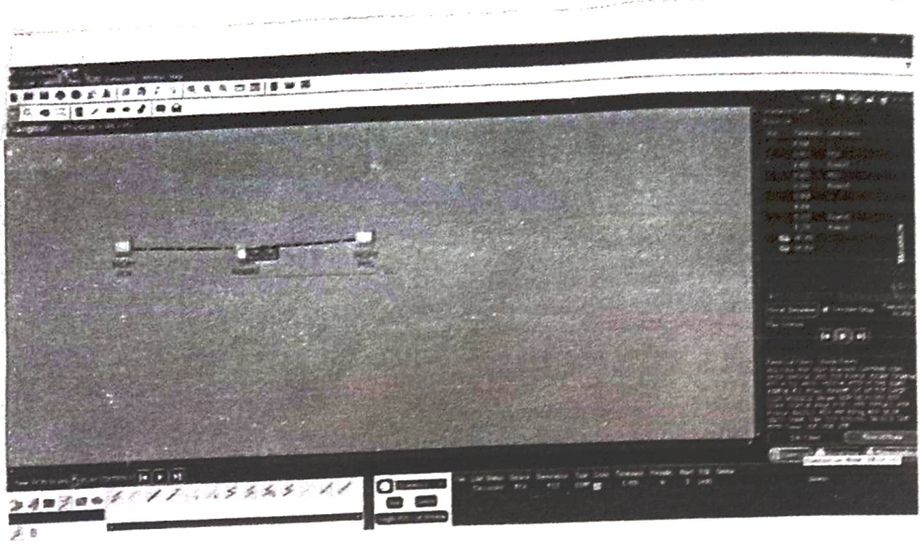
IP address {IP address} {subnet mask}

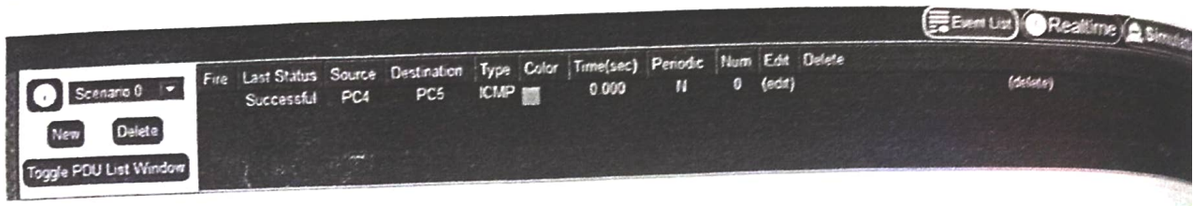
no shut down

exit

Result:

Therefore, the subnetting on cisco packet tracer simulator was executed successfully.






PC0

Physical | Config | Desktop | Programming | Attributes

**Command Prompt**

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 208.67.220.220

Pinging 208.67.220.220 with 32 bytes of data:

Request timed out.
Reply from 208.67.220.220: bytes=32 time=1ms TTL=127
Reply from 208.67.220.220: bytes=32 time=1ms TTL=127
Reply from 208.67.220.220: bytes=32 time=1ms TTL=127

Ping statistics for 208.67.220.220:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>
```

PC0

Physical | Config | Desktop | Programming | Attributes

**Command Prompt**

```
C:\>ping cisco.com

Pinging 208.67.220.220 with 32 bytes of data:

Reply from 208.67.220.220: bytes=32 time=9ms TTL=127
Reply from 208.67.220.220: bytes=32 time=1ms TTL=127
Reply from 208.67.220.220: bytes=32 time=1ms TTL=127
Reply from 208.67.220.220: bytes=32 time=2ms TTL=127

Ping statistics for 208.67.220.220:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 9ms, Average = 3ms
```

10.

a) Internet working routes in CISCO Packet tracer.

Aim:

To design and configure a simple internetwork using a router in CISCO packet tracer and Fast connecting two PCs.

Procedure:

Step 1:

Configure router:

* select the routes and open CLI
* press enter to start configure Router.
* Type enable to activate the privileges mode.

Step 2:

* Assign IP address to every PC in the network.
* Select the PC, go to desktop and select IP configure and assign IP address Default gateway.
* Assign's the default gateway of PC's 192.16.8.10.1
* Assign the default gateway of Pc's 192.168.20.1.

Steps:

* connect fast ethernet a part of with fast ethernet 0/0 part of router, using a cop port straight through cable.

* Connect Fast ethernet 0 power fast ethernet all port of row as using a copper straight through cable

PC1

Physical | Config | Desktop | Programming | Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 20.0.0.1

Pinging 20.0.0.1 with 32 bytes of data:

Reply from 20.0.0.1: bytes=32 time<1ms TTL=255
Reply from 20.0.0.1: bytes=32 time<1ms TTL=255
Reply from 20.0.0.1: bytes=32 time<1ms TTL=255
Reply from 20.0.0.1: bytes=32 time<1ms TTL=255

Ping statistics for 20.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.0.0.2

Pinging 10.0.0.2 with 32 bytes of data:

Reply from 10.0.0.2: bytes=32 time=19ms TTL=126
Reply from 10.0.0.2: bytes=32 time=19ms TTL=126
Reply from 10.0.0.2: bytes=32 time=18ms TTL=126
Reply from 10.0.0.2: bytes=32 time=19ms TTL=126

Ping statistics for 10.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 18ms, Maximum = 19ms, Average = 18ms

C:\>tracert 10.0.0.2

Tracing route to 10.0.0.2 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      20.0.0.1
  2    11 ms     13 ms     12 ms     192.168.1.254
  3    9 ms      5 ms      5 ms      10.0.0.2

Trace complete.

C:\>
```

Top

```
C:\> tracert 20.0.0.2

Tracing route to 20.0.0.2 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      10.0.0.1
  2    6 ms      8 ms      7 ms      192.168.1.253
  3    5 ms      10 ms     7 ms      20.0.0.2

Trace complete.
```

Result:

Therefore, the Internet work
routers in CISCO packet tracer.

10.5) Design and configure an Internet using wireless router, DHCP Server and Internet cloud.

Aim:

To design and configure an Internet using a wireless router DHCP server and Internet cloud in cisco packet tracer.

Procedure:

* Open cisco packet tracer and create required network topology.

* Connect pc: laptop wireless router, cable, modem, & Internet cloud and server.

* configure the laptop to the wireless router network and pc to DHCP

* configure the server with IP 208.88.x and enable DHCP and DNS

* Verify connectivity using IP config and Ping cisco.com

PC0

Physical | Config | Desktop | Programming | Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 30.0.0.100

Pinging 30.0.0.100 with 32 bytes of data:

Request timed out.
Reply from 30.0.0.100: bytes=32 time=2ms TTL=126
Reply from 30.0.0.100: bytes=32 time=18ms TTL=126
Reply from 30.0.0.100: bytes=32 time=24ms TTL=126

Ping statistics for 30.0.0.100:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 24ms, Average = 14ms

C:\>tracert 30.0.0.100

Tracing route to 30.0.0.100 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      10.0.0.1
  2    1 ms      0 ms      0 ms      40.0.0.2
  3    3 ms      2 ms      2 ms      20.0.0.2
  4    2 ms      5 ms      0 ms      30.0.0.100

Trace complete.

C:\>
```

```
C:\>tracert 30.0.0.100

Tracing route to 30.0.0.100 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      10.0.0.1
  2    3 ms      1 ms      1 ms      20.0.0.2
  3    0 ms      0 ms      1 ms      30.0.0.100

Trace complete.
```

```
C:\>tracert 30.0.0.100

Tracing route to 30.0.0.100 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      10.0.0.1
  2    1 ms      7 ms      4 ms      40.0.0.2
  3    9 ms      7 ms      1 ms      20.0.0.2
  4    2 ms      0 ms      4 ms      30.0.0.100

Trace complete.

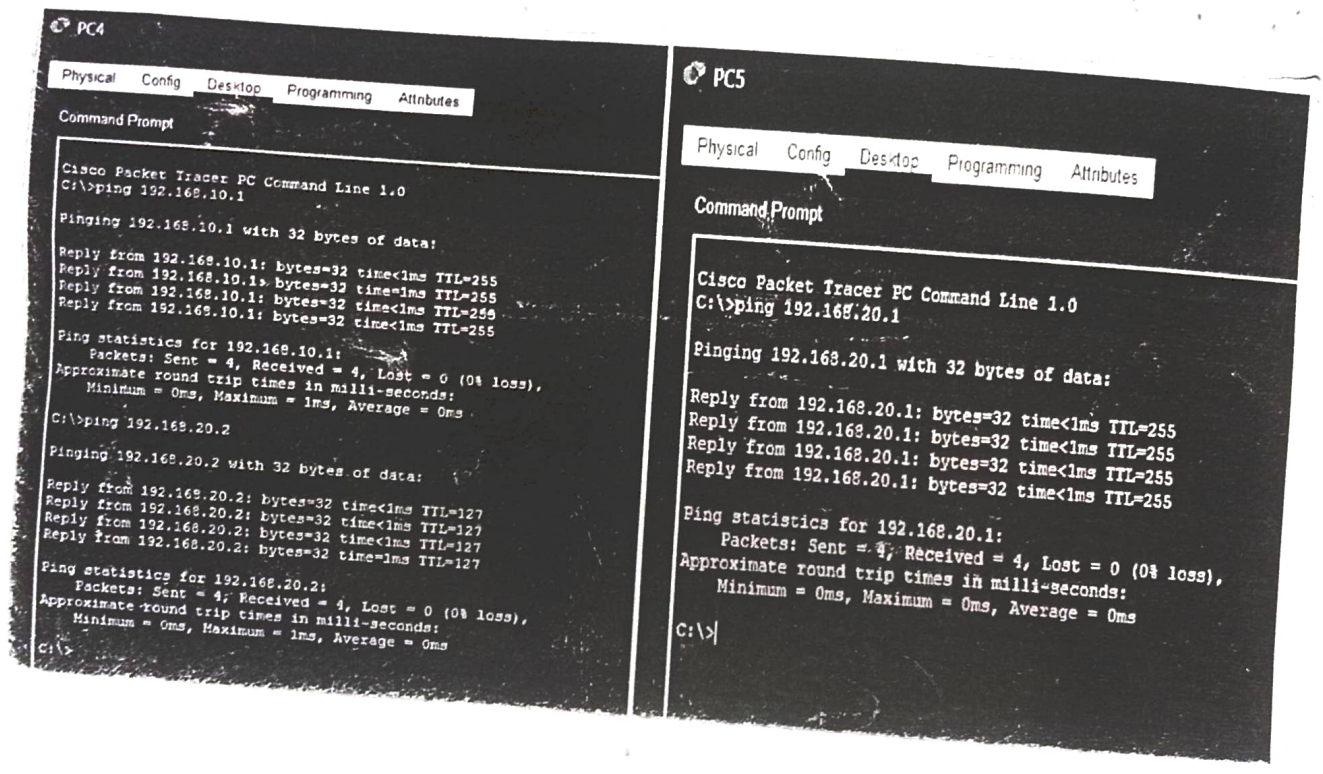
C:\>
```

4) a) Simulate static routing configuration
Exp using cisco packet tracer

Aim:
To simulate a static routing configure using cisco packet tracer.

Procedure:
* Open cisco packet tracer and create the given network topology.
* Connect 3 routers and PC's using suitable cable
* Connect IP address for router and PC's
* Enable and router interface using no shutdown
* Configure static routes router = using ip route.
* Set main and backup routers using different administrative distance values
* Test connectivity.
* User tracer verify the routing path
* Disconnect the main link and verify that the backup router work.

a link and router again to verify RIP. select the alternate route.




Result:

Therefore, the simulate static routing configuration using cisco packet tracer was executed successfully.

4b) Simulate RIP using cisco packet tracer

Aim:
To simulate RIP using 3 routers and 2 pc's cisco packet tracer.

Procedure:
* Create the topology with 3 routers and 2pc's
* connect the device using suitable cables.
* Assign the given IP address to both PC's
* Configure fast ethernet and serial interface of all routers.
* Enable interface using no shutdown.
* Set clock rate of 64000 and bandwidth 64n for serial interface.
* Enable RIP on all routers using the cisco
* Set clock rate of 4010 and bandwidth 64n.
* Add directly connected network using network command

* Test connectivity using Ping 20.0.0.1
* Disconnected the direct cable between router 2 Link and router again to verify RIP select the alternate route.

Result:

Thus, the simulation of RIP using 3 routers 2PCLs. cisco packet tracer has been successfully done.

Exp: 12.

End- End communication at Transport Layer

a) Implement echo client server using TCP/UDP Socket

b) Implement a chat program: using Socket program.

TCP:

```
import socket
import threading

# server --
def start_server()
    server = socket.socket (socket.AF_INET,
     socket.SOCK_STREAM)
    server.bind (('127.0.0.1', 5000))
    server.listen(1)
print ("[SERVER] listening on 127.0.0.1:5000")
conn, addr = server.accept()
print(f"[SERVER] connected by {addr}")
while true:
    data = conn.recv(1024).decode()
    if not data or data.lower()= 'quit':
        print (SERVER) connection closed by
            client")
        break
    print(f"(Server) connection
        closed by client")
conn.close()
server.close().
```

```
# client...
def start_client():
    client = socket.socket(Socket.AF_INET,
                           Socket.SOCK_STREAM)
    client.connect(('127.0.0.1', 5000))
    print("[client] connected to server")
    while true:
        msg = input("Enter message (quit/exit)")
        client.send(msg.encode())

if msg.lower() == 'quit':
    break

response = client.connect(('127.0.0.1', 5000))
print(f"[Server Response] {response}")
client.close()

# --Main--
if name == "__main__"
    server_thread = threading.Thread(target=start
        server, daemon=true)
    server_thread.start()
    start_client()
```

Output:

[server] listening on 127.0.0.1:5000 [client]
Connected to server.
hello
[Server] received hello
[Server Response] Server received 'hello'
Enter message (or 'quit' or 'exit')

UDP :

```
Import socket
Import threading
# server --
def udp_server()
    s = socket.socket(Socket.AF_INET.Socket.
        SOCKET.DGRAM).
    s.bind ('127.0.0.1',12000).
    print ("[SERVER] Running on 127.0.0.1:12000")

while true:
    data.addr = s.recvfrom(1024)
    msg = data.decode()
    if msg.lower() == 'quit'
        print ("[server] client {addr}
enable the connection").
    break
    printf ('SERVER] Received from {addr} {msg}')
    s.sendto (data, addr).
s.close()
print ("[SERVER] STOPPED")

# client --
def udp_client()
    c = socket.socket (Socket.AF_INET, Socket.
        Sock_PGRAM)
    server = ('127.0.0.1', 12000)
    while true:
        msg = Input ("Enter message ")
```

```
        c.send to (msg.encode(), serv)
        if msg.lower() == "quit":
            print ('[client] closing connection...')
            break.

    data, _ = c.recvfrom (1024).
    print ("Echo from server", data.decode())
    c.close()
    print ("[CLIENT] stopped")

#--- Main ---
if __name__ == '__main__':
    threading.Thread (target = udp_server
```

Output:

```
Enter a message: [server] Running on
127.0.0.1 12000 hi
[server] Received from ('127.0.0.1', 49670)
Echo from server: hi
Enter message (or quit to stop) how are
you [server]
```

Result:

Thus above two ~~program~~ was executed successfully.

EXP : 13

Implement own PING Program

AIM:

To Implement my own Ping program.

Code:

```
import os
import platform
import subprocess
def ping(host="google.com", count=4):
    param = "-n" if platform.system().lower()=="windows" else "-c"
    command = ["ping", param, str(count), host]
    try:
        output = subprocess.check_output(command, universal_newlines=True)
        print(output)
    except Exception as e:
        print(f"Ping failed: {e}")
if __name__ == "__main__":
    ping("8.8.8.8")
```

Output:

```
Pinging 8.8.8.8 with 32 bytes of data
Replying from 8.8.8.8: bytes=32 time=6ms
```

Replying from 8.8.8.8: bytes=32 time = 9ms

Replying from 8.8.8.8: bytes=32 time = 5ms

Replying from 8.8.8.8: bytes=32 time = 9ms

Ping statistics for 8.8.8.8.

Packets. sent = 4 Received = 4 Lost = 0 (0%)

Approximate round trip times in milli-sec

Minimum. 5ms, Maximum = 9ms,

Average = 7 ms

Result:

This the ping program is worked and executed successfully.

Exp: 14

Raw Socket to Implement packet sniffing.

Aim:

To write a code using raw socket for packet sniffing.

code:

```
from scapy.all import sniff,
from scapy.layers.inet import IP, TCP, UDP,
    ICMP
def packet_callback(packet):
    if IP in packet:
        ip_layer = packet[IP]
        protocol = ip_layer.proto
        src_ip = ip_layer.src
        dst_ip = ip_layer.dst
        protocol_name = ""
        if protocol==1:
            protocol_name = "ICMP"
        elif protocol==6:
            protocol_name = "TCP"
        elif
            protocol==17:
            protocol_name = "UDP"
        else:
            protocol_name = "Unknown
                Protocol"
print(f"Protocol: {protocol_name}")
print(f"Source IP: {src_ip}")
print("-"*50)
```

Out

sniff (iface = "wlfi", prn. packet - callback,
filter = "ip", store = 0)

Output.

C:/users/Sathu/Downloading."python packet sniffer.py"

Protocol: TCP.

Sourca IP: 57.15.141.117.

Destination IP: 192.168.1.6

---

Protocol: TCP.

Source IP: 192.168.1.8

Destination IP: 57.155.141.117

---

Protocol: UDP

Sourca IP: 192.168.1.16.

Sourca IP: 192.168.1.1

Destination IP: 192.168.1.1

---

Protocol: UDP

Sourca IP: 192.168.1.1

Destination IP: 192.168.1.6

Result:

Thus the raw sockets is used to Implement packet sniffing successfully.